



Emphasis on Digital Innovations and Sustainability in the Hospitality Industry: An Explorative Study Based on a Survey of Restaurant Guests

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Received: 01-10-2025

Revised: 03-03-2025

Accepted: 03-17-2025

Citation: Ludin, D., Drautz, S., & Urban, S. (2025). Emphasis on digital innovations and sustainability in the hospitality industry: An explorative study based on a survey of restaurant guests. *Tour. Spectr. Div. Dyn.*, 2(1), 1–6. <https://doi.org/10.56578/tsdd020101>.



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Abstract: The impact of digitization has gradually reached the arena of hospitality industry and is gaining more importance, especially after pandemic. Digital menus, online reservations, and cashless payments are just a few examples which restaurant guests are used to these days. The concept of sustainability has drastically transformed the operation of the hospitality industry: organic products, recycling, reuse, respectful handling of food, etc. The objective of this study was to analyze whether the sustainability and/or digital innovations in a restaurant were decisive for the choice of a restaurant. The method applied in this study was based on the state of research and knowledge of the topic; the research design could be described as “an interplay of literature search and survey”. The findings demonstrated that sustainability was important for restaurant guests, but not decisive for selecting a restaurant. Neither attending to sustainable aspects nor the implementation of digital innovations had a strong impact on the choice of a dining place. Yet, sustainability was weighed slightly more heavily than digital innovations. For restaurant owners, the results revealed that sustainability aspects and digital innovations could be crucial in the running of daily business. To conclude, the study contributes to the theoretical discourse on the importance of digital innovations and sustainability in the hospitality industry and points to the direction for further development of the existing theories.

Keywords: Digital innovations; Sustainability; Hospitality industry; Restaurants; Guests

1. Introduction

The operation of the hospitality industry has benefitted from the implementation of digitization and its increasing importance, especially after pandemic times. To name a few examples of affected areas in restaurants, guests can now enjoy digital menus, online reservations, and cashless payments. Furthermore, the growing awareness of sustainability has impacted the hospitality industry and brought forth organic products, recycling, reuse, and respectful handling of food. Therefore, the aim of this study is to analyze whether the concept of sustainability and/or utilization of digital innovations are decisive for determining the choice of a restaurant.

2. Literature Review

2.1 Method and Focus of the Literature Review

The literature review integrated the present study into the current state of research (Bramer et al., 2018). Deriving specific research questions (RQs), it encompassed the indexing and processing of the findings documented in the literature on the scientific fields of “digital innovations”, “sustainability”, and “hospitality industry” (Snyder, 2019). Areas aligned with the scope of the study were identified and outlined with a curated selection of relevant materials. These materials covered articles from the past five years (with a few articles that are not in this spectrum) and online resources. Utilizing the online database “Resurchify” (<https://www.resurchify.com/ranking>), pertinent articles from relevant journals were screened. The focus was on

the development of explanatory approaches as well as the operationalization of relevant variables for empirical investigation. This analytical approach followed a thematic sequence of steps, starting from defining the problem and leading to the emergence of central questions that shaped the focus of the study (Eco, 2015).

2.2 Digital Innovations and Sustainability in the Hospitality Industry

The last five years in the hospitality industry have been very much shaped by the COVID-19 pandemic. Restaurants were facing problems because either people were not allowed to visit restaurants, or in times, when it was possible, many people did not want to expose themselves to any risk and preferred to eat at home, or to have food delivered, or to pick it up (Brewer & Sebbby, 2021; Yang et al., 2020). Many restaurants therefore had to revise their strategies and had to find original ideas to reach their customers. It became clear that digital innovations are also necessary in the hospitality industry. Many restaurants have introduced innovations during these times, such as ordering food online to be delivered at home or to the front door (Seetharaman, 2020). They also have offered orderings for pick-up in to-go food boxes. Apparently, quality, price, and hygiene are decisive points for customers when choosing the right provider (Khan et al., 2023). When people at home did not have much variety in life due to lockdowns, these digital innovations helped to provide options. Therefore, restaurant guests react now more positively to digital innovations than before the pandemic (Yap et al., 2021).

Digitization offers added value not only for guests, but also for the restaurants themselves. Many restaurants have invested more in digitization, what has helped them to differentiate themselves from other competitors (Ruiz-Molina et al., 2014) and to be more attractive to guests. Communication and information technologies could also increase productivity, reduce errors, and contribute to better organization (Sigala, 2003). The COVID-19 pandemic has accelerated the adoption of digital technologies in the hospitality sector. Due to government-mandated closures, hospitality entrepreneurs in Germany have developed new digital business models and utilized online platforms as well as digital marketing. In this way, restaurateurs and managers were able to adapt their business operations during the lockdowns (Ludin et al., 2022). In the hospitality industry, service robots are used (Li & Wang, 2022); these robots are perceived by customers as an innovative step towards digitalization (Wirtz et al., 2018). For example, customers in Korea can order coffee from a robot barista (Kim et al., 2022). So far, however, these robots have only been on the rise in Asian countries (Lu et al., 2019). Another trend, self-service technology, is also noticeable and has already been known by large fast-food chains. This trend has now been taken over by traditional restaurants. For instance, there are digital menus that can be scanned via quick response (QR) code and can be placed directly on a tablet (Ozturkcan & Kitapci, 2023). Furthermore, the Internet and social media play an increasingly important role in the hospitality industry. People love to go out to eat and rely on the reviews concerning restaurants on the Internet (Rhee et al., 2016). There are more food blogs that recommend both dishes and restaurants. The younger generations are eager to try out new things when inspired by these blogs. That is why it is important for restaurants to be mentioned by food bloggers (Lee et al., 2021). In general, it must be mentioned that it is easier for generations Y and Z to get used to the digitization of restaurants owing to the immersion of technology in their growing up process. That is why digital solutions, such as digital reservations or menus, are always welcomed and accepted by these young people (Ägnes, 2022).

In recent years, another trend has become more prevailing: there are more people who are concerned with the sustainability of restaurants. The hospitality industry faces growing consumer demand for sustainable operations, with market data indicating a distinct guest preference for eco-conscious restaurants. Industry reports suggested that customers preferred to stay in hotels that care about the environment. They attached importance to the fact that some restaurants also share these values (Khatter et al., 2019). Many sustainable restaurants have followed this trend to stand out from competitors, as this is a promising way to stand out from the crowd (Kim & Hall, 2020; Rather et al., 2022). Because of the use of sustainable resources and sustainable work equipment, a restaurant can advertise by highlighting these features via the social media. The marketing of sustainability on the social media would, in turn, demand a higher price from the customers subsequently (Chuah et al., 2020).

Since a conscious and regional diet is important to restaurant guests (Winterstein & Habisch, 2021), more restaurants advertise with regional, organic, and fresh products. For these dishes, guests are willing to pay a higher price (Chuah et al., 2020). Another point for attracting customers is sustainable packaging, further developed during the COVID-19 pandemic and the expansion of the delivery service (Eger et al., 2021). More and more restaurants are offering reusable packaging or coffee mugs (Keller et al., 2021). This could reduce costly and very time-consuming separation processes of the various packaging methods and is particularly beneficial for the environment. In addition, there emerges a new concept of “too-good-to-go” in many cities, with explicit advantages for restaurants and customers. Here, restaurants could sell the leftover food to guests for a small price. The restaurants no longer have to throw away food and can still generate a small profit via this means.

2.3 Research Gaps and Research Questions

To date, only a few studies have addressed the necessity of transforming digital technology and sustainability

(Batat, 2021; Brewer & Sebby, 2021; Kim & Hall, 2020; Rather et al., 2022; Wilkesmann & Wilkesmann, 2020) in the gastronomy sector. To contribute to this field of research, the following RQs formed the basis for the design of a quantitative customer survey and data analysis:

RQ 1: How important the sustainability of a restaurant is for guests; is it decisive for the choice of a restaurant?

RQ 2: How important digital innovations in a restaurant are for guests; are they decisive for the selection of a restaurant?

3. Method

The method applied here was selected based on the state of research and knowledge of the topic. Therefore, the research design can be described as an interplay of literature research and a survey. With the help of the survey, new perceptions were explored and new insights could subsequently be derived. Before disseminating the questionnaire, a pretest consisting of a thorough walkthrough of the questionnaire with a peer researcher was performed to enhance the reliability of the survey instrument. Instead of utilizing items from prior studies, the questionnaire was custom-designed and derived directly from the theoretical frameworks established in the literature review. For data collection, the quantitative survey was conducted from 24th April 2023 to 8th May 2023 with the help of a digital form created via Survio (<https://www.survio.com>). The link to the questionnaire was distributed to restaurant guests via a QR code in the following six restaurants in Heilbronn:

- Burgerheart (<https://heilbronn.burgerheart.com>)
- Charivari (<https://www.charivari-hn.de>)
- Lehnerns Wirtshaus (<https://heilbronn.lehnerns-wirtshaus.de>)
- L'OSTERIA (<https://losteria.net/de/restaurants/restaurant/heilbronn>)
- MoschMosch (<https://www.moschmosch.com/restaurants-heilbronn.php>)
- VELO (<https://velo-heilbronn.de>)

The adoption of a QR code enables restaurant guests to access the survey website in a prompt manner. Upon scanning the code, the user is automatically redirected to the document. The utilization of the QR code was intended to reduce the time required by respondents, thereby increasing the participation rate. In addition, the questionnaire was forwarded online to achieve greater reach and ultimately 96 people took part in the survey. Descriptive statistical procedures were used for data analysis and interpretation.

4. Key Results

Sustainability has become more and more important to guests in recent years in the hospitality sector. It was discovered that 56% of the respondents considered the sustainability of a restaurant important, 18% very important, while 23% considered it less important and only 3% considered sustainability to be unimportant. This meant that almost three quarters (74%) of the respondents considered it important or very important for restaurants to pay attention to sustainability. For the surveyed guests, “careful planning to reduce food waste” (85%), “use of seasonal and regional products” (85%), “use of recyclable and biodegradable packaging” (61%), and “use of environmentally-friendly cleaning products” (32%) were essential to promoting sustainability in a restaurant. Then, the survey asked whether guests would be willing to pay a higher price in a restaurant for sustainable products or practices. 70% of the respondents were willing to pay a surcharge for sustainable products/management, if it was not too expensive. 18% would even pay a higher price in any case.

The next question was to enquire which digital innovations were decisive for customers when choosing a restaurant. With 71% digital payment being the most important form, followed by online reservation option of 53% and digital menu of 41%. This illustrated that digital payment was particularly popular among customers since the COVID-19 pandemic, as contactless payment has gained enormous importance since then. The digital menu, which can be scanned through the QR code, has probably become part of everyday life after the pandemic. It asked whether people would try a new restaurant “that uses digital tools” on a recommendation. Here, 47% of the respondents answered “maybe”, 42% answered “yes, definitely”, and 11% replied with “no, rather not”. Then it asked how digital innovations could help restaurants to be more sustainable. “Optimizing ordering and warehousing” seems to be the most important (85%), followed by “monitoring resource consumption and identifying savings potential” (60%) and “facilitating collaboration with other companies to promote the circular economy” (59%). In the next step, guests were asked whether they would go to restaurants more frequently if (a) “restaurants would pay more attention to sustainability”; (b) “restaurants would use more digital tools”; or (c) “no preference”. Results suggested that 62% of the respondents had “no preference”, while 20% would visit restaurants that paid more attention to sustainability, 12% would visit restaurants that used more digital tools. Since most respondents did not indicate any preferences, it was concluded that sustainability and digital innovations did not have a large impact on the frequency of restaurant visits. However, a small gradation can be made between the

two trends. Sustainability has a small impact of 20%, higher than the 12% of digital innovations. The trend of sustainability is therefore slightly stronger than the trend towards digital innovations.

5. Conclusion

5.1. Discussion

After the literature review, it became evident that the findings of the present study contributed to the expansion of existing theories. This study facilitated the development of an in-depth understanding of the significance of digital innovations and sustainability in the hospitality industry. It enriched the theoretical discourse on the roles of digital innovations and sustainability in the hospitality industry and provided points of reference for further development of the existing theories. Empirical findings revealed that almost three thirds of the guests considered it important or even very important that restaurants emphasized sustainability. The reduction of food waste and the utilization of seasonal and regional products could be critical to most of the restaurant guests. “Optimizing ordering and warehousing” seemed to be most important when it helped restaurants to be more sustainable through digital innovations. Most of the respondents would pay a higher price for sustainable products/management in a restaurant (if the price was reasonable). The results concluded that sustainability was crucial for restaurant guests, but not decisive for the choice of a restaurant (RQ 1). Neither paying attention to sustainable aspects nor the implementation of digital innovations had a strong influence on choosing a restaurant. However, sustainability was weighed slightly more heavily than digital innovations; maybe because sustainability directly dealt with core values, tangible quality, and emotional connections, whereas digitalization was often perceived merely as a convenience tool. In general, digital innovations, especially digital (cashless) payments, offered an indispensable option to restaurant guests. Yet, the recommendation of a restaurant due to digital innovations was not relevant for most respondents (RQ 2).

5.2. Limitations and Future Research

The survey presented was exploratory in nature and therefore not representative. Due to the relatively small sample size and its geographic concentration within a single city in Germany, the generalizability of the findings was constrained. To gain a thorough understanding, it is necessary to undertake a larger-scale survey involving a greater number of restaurant guests involving broader geographic coverage. Nevertheless, the number of samples is sufficient to ensure a qualitative result. Future research could examine the requisite “future skills” that would enable gastronomic enterprises to advance sustainability agendas and navigate digital transformations. Besides, analysing the strategic impact of loyal customers as primary drivers of a restaurant’s innovative capacity, as well as its inclination toward sustainable and digital adaptation, warrants further empirical investigation.

5.3. Practical Implications

For restaurant owners and managers, the results encouraged them to handle sustainability aspects and digital innovations in daily business. Especially in small and medium-sized restaurants, the following digital tools may be implemented most effectively without creating excessive operational costs: online menus, online ordering, online shops, online tastings, online cooking classes, online marketing, and social media marketing. In the realm of sustainability, for instance, the following measures hardly incur any additional costs: cooperation with local producers, the use of seasonal menus, and analogous waste-reduction concepts such as the “nose-to-tail” principle (the holistic utilization of food).

Author Contributions

Conceptualization, D.L.; methodology, D.L., S.D., and S.U.; validation, D.L.; investigation, D.L., S.D., and S.U.; data curation, D.L.; writing—original draft preparation, D.L., S.D., and S.U.; writing—review and editing, D.L.; visualization, D.L.; supervision, D.L.; project administration, D.L. All authors have read and agreed to the published version of the manuscript.

Data Availability

The data used to support the research findings are available from the corresponding author upon request.

Conflicts of Interest

The authors declare no conflicts of interest.

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